Essays in Causal Inference with Panel Data

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Modern causal methods with panel data for empirical research

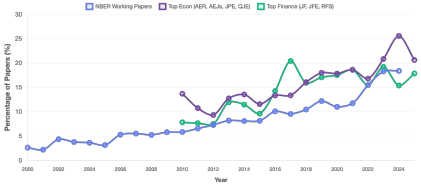
- A rapidly expanding literature has emerged on methods for estimating causal effects in *panel data* contexts.

- Recent meta-analysis has documented the growing use of *difference-in-differences* (DiD), *event-study*, *triple-differences*, and *fixed-effects* methods in NBER WP and leading economics journals, reflecting substantial progress in the understanding/adoption of such methods.[1].

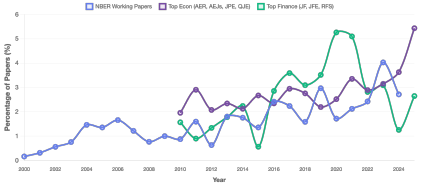
[1] See Goldsmith-Pinkham (2024) for a complete discussion.

Current trends in empirical research

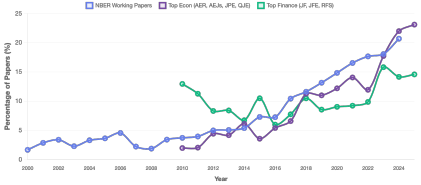
The following figures show these trends at: paulgp.com/econlit-pipeline



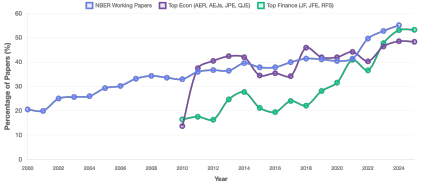
Difference-in-Differences



Triple Differences



Event Studies



Fixed Effects

Challenges and Opportunities

- When PTA is *questionable*, practitioners use placebo comparison groups, but the econometrics of *triple difference* designs remains poorly understood.
- When treatment is *continuous*, there is little to no work on *heterogeneity analysis* in DiD.
- *Latent factor models* offer a richer alternative to TWFE, however existing methods impose linearity which may be restrictive.

My proposal:

- **Triple Differences Designs**.
- **Heterogeneous Treatment Effects** with continuous treatments.
- **Reweighting and Matrix Completion** in nonlinear panel models.

Roadmap

1. Better Understanding Triple Differences Estimators
with Pedro H.C. Sant'Anna

2. Heterogeneous Moderating Effects in Difference-in-Differences with Continuous Treatment

3. Flexible Reweighting and Matrix Completion Methods in Nonlinear Causal Panel Data

Better Understanding Triple Differences Estimators

with Pedro H.C. Sant'Anna

What is Triple Differences?

Example: Gruber (1994)

- **Program**: Effect of *mandated maternity benefits* (1975–1978) on wages and labor supply of women of childbearing age.
- **Problem**: DiD-PTA may fail if state-specific shocks correlate with mandate adoption timing.
- **DDD to the rescue**:
  - *Treated*: married women aged 20–40 →affected by mandates.
  - *Comparison*: women over 40 and single men 20–40 →unaffected by the policy but exposed to the same state-specific shocks.

TABLE 3—DDD ESTIMATES OF THE IMPACT OF STATE MANDATES ON HOURLY WAGES

Location/year	Before law change	After law change	Time difference for location
A. *Treatment Individuals: Married Women, 20–40 Years Old:*			
Experimental states	1.547 (0.012) [1,400]	1.513 (0.012) [1,496]	−0.034 (0.017)
Nonexperimental states	1.369 (0.010) [1,488]	1.397 (0.010) [1,640]	0.028 (0.014)
Location difference at a point in time:	0.178 (0.016)	0.116 (0.015)	
Difference-in-difference:	−0.062 (0.022)		
B. *Control Group: Over 40 and Single Males 20–40:*			
Experimental states	1.759 (0.007) [5,624]	1.748 (0.007) [5,407]	−0.011 (0.010)
Nonexperimental states	1.630 (0.007) [4,959]	1.627 (0.007) [4,928]	−0.003 (0.010)
Location difference at a point in time:	0.129 (0.010)	0.121 (0.010)	
Difference-in-difference:	−0.008: (0.014)		
DDD:	**−0.054 (0.026)**		

Notes: Cells contain mean log hourly wage for the group identified. Standard errors are given in parentheses; sample sizes are given in square brackets. Years before/after law change, and experimental/nonexperimental states, are defined in the text. Difference-in-difference-in-difference (DDD) is the difference-in-difference from the upper panel minus that in the lower panel.

Fig 1: DDD Estimates - Wage Effects

DDD designs address this issue by finding *within-group comparison units* that **are not affected** by the treatment but **are exposed** to a time-varying confounder.

The appeal of DDD compared to DiD

In DDD we are able to use all the information to control for location-specific trends and partition-specific trends, which otherwise would yield questionable results using DiD.

	No Eligible	Eligible
Treated	(EXP, M/Older W)	(EXP, 20-40 yo W)
Control	(Non-EXP, M/Older W)	(Non-EXP, 20-40 yo W)

DDD mechanics

Contribution

- Study identification, estimation, and inference for ATT and event-study-type parameters in DDD settings.

- Provide results for several setups:
  - With and without covariates.
  - Block treatment and staggered treatment adoption.

- Challenge common DDD practices:
  - *Standard regression-based* DDD can impose restrictive assumptions and lead to *bias*.
  - With covariates, absent additional restrictions, DDD estimators generally **cannot** be written as *"difference of two DiDs."*
  - In staggered DDD, pooling all *not-yet-treated* units as controls can induce substantial *bias*, which is in sharp contrast to staggered DiD setups.

Why is this relevant for empirical practices?

In the classical case, one could recover the ATT in DDD either (i) by a 3WFE regression or (2) by a difference of 2 DiDs.

What happens when *covariates* play an important role?

Empirical-Calibrated Monte Carlo Simulation

- Let's start with some simulations first based on Gruber (1994).
- Using an **empirically-calibrated** simulation design, we combine:
  - *Wasserstein GANs* trained on CPS March Supplement data to generate covariates and outcomes closely resemble the actual data (Athey et al., 2024).
  - Use *estimated coefficients* from real data to compute a multinomial logit propensity score-based (S, Q) assignment (Busso et al., 2014).

We draw 5k Monte Carlo experiments and each simulated sample contains $n \approx 27{,}575$ with:

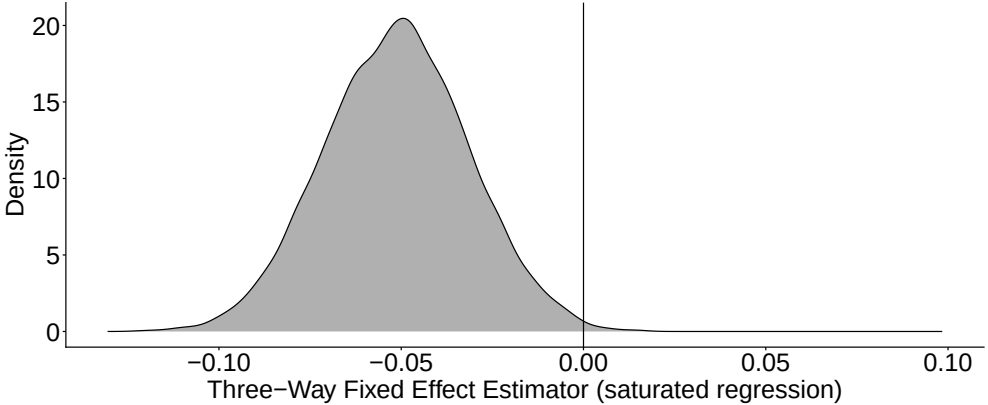
- **State indicator**: $S_i \in \{2, \infty\}$ (experimental state vs. non-experimental states)
- **Eligibility indicator**: $Q_i \in \{0, 1\}$ (women over 40 and single males 20-40 vs. married women 20-40 years old)
- **Time indicator**: $t \in \{1, 2\}$ (before vs. after law change)

MC Simulations Design

Challenging folks' wisdom: 3WFE?

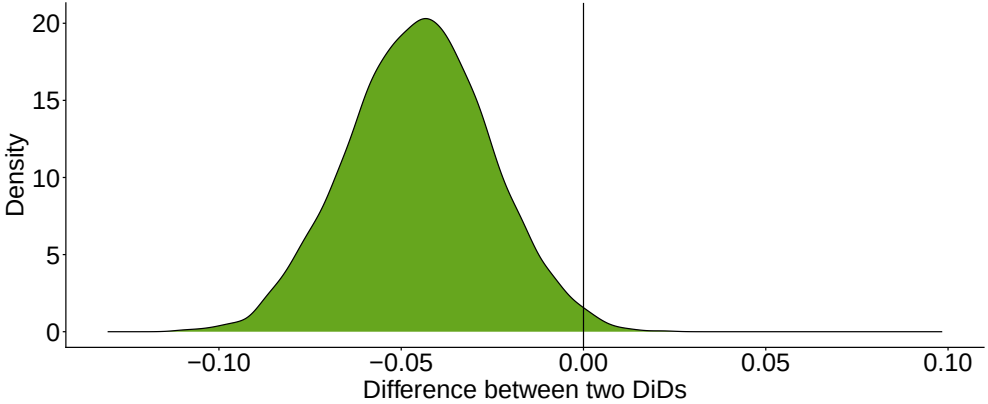
Add covariates linearly and S, Q, T interactions in a 3WFE regression:

$$Y_{i,t} = \beta_0 + \beta_1 1\{S=2\} + \beta_2 1\{Q=1\} + \beta_3 1\{T=2\} + \beta_4 1\{S=2\}1\{Q=1\}$$
$$+ \beta_5 1\{S=2\}1\{T=2\} + \beta_6 1\{Q=1\}1\{T=2\} + \beta_{3wfe} 1\{S=2\}1\{Q=1\}\{T=2\} + X\Gamma + u_{i,t}$$



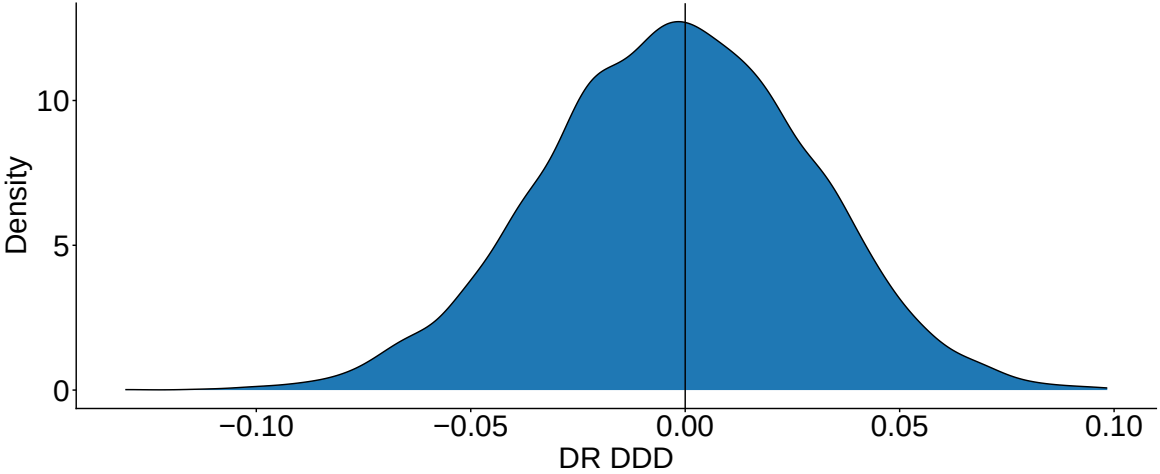
Challenging folks' wisdom: Difference of 2 DiDs?

DDD estimates as the difference between two DiD estimates: one DiD using the subset with $S = 2$ and considering units treated if $Q = 1$, and another DiD using the subset with $S = \infty$ and considering units "treated" if $Q = 1$.

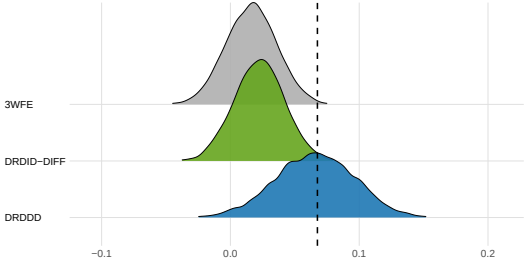


Can we do better?

Challenging folks' wisdom: DR DDD!



Challenging folks' wisdom: DR DDD!



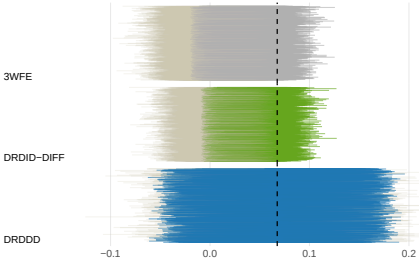


Table 1: Comparison of estimators across metrics

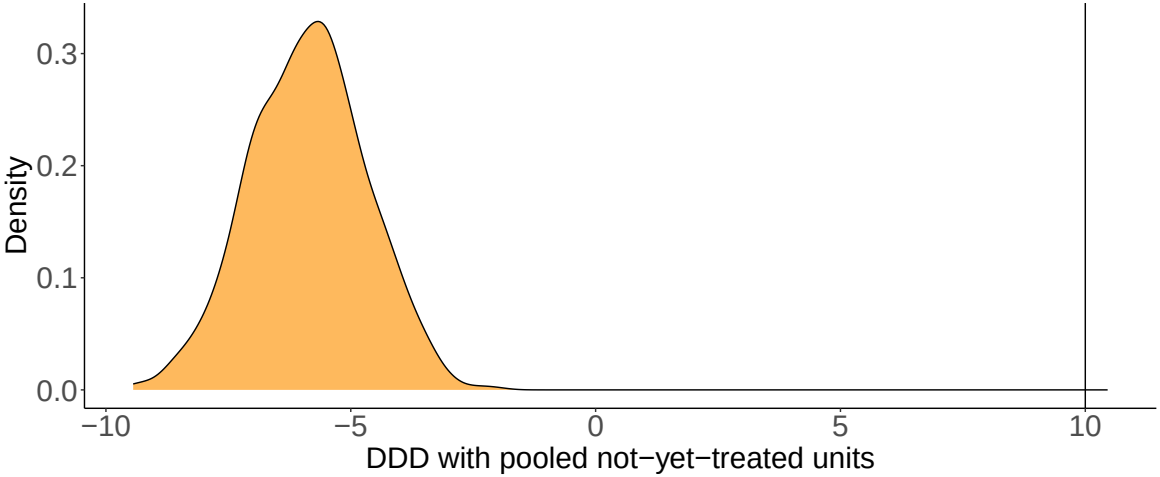
Estimator	Relative Bias	Relative RMSE	Coverage	Relative ALCI
DRDDD	1.00	1.00	0.948	1.00
3WFE	25.65	1.78	0.342	0.715
DRDID-DIFF	22.55	1.59	0.358	0.627

What about staggered adoption settings in DDD?

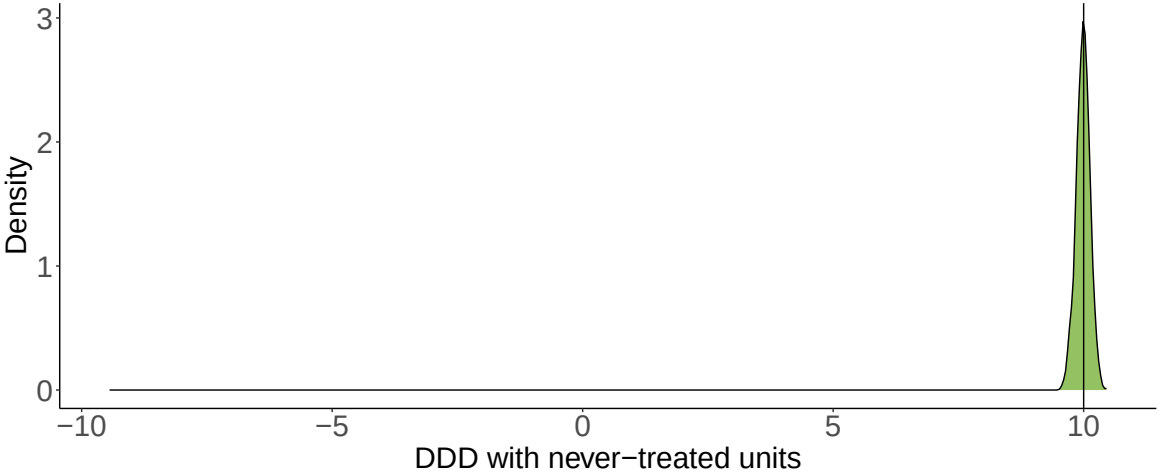
Which comparison groups should we use? *never-treated* or *not-yet-treated* units?

MC simulation design

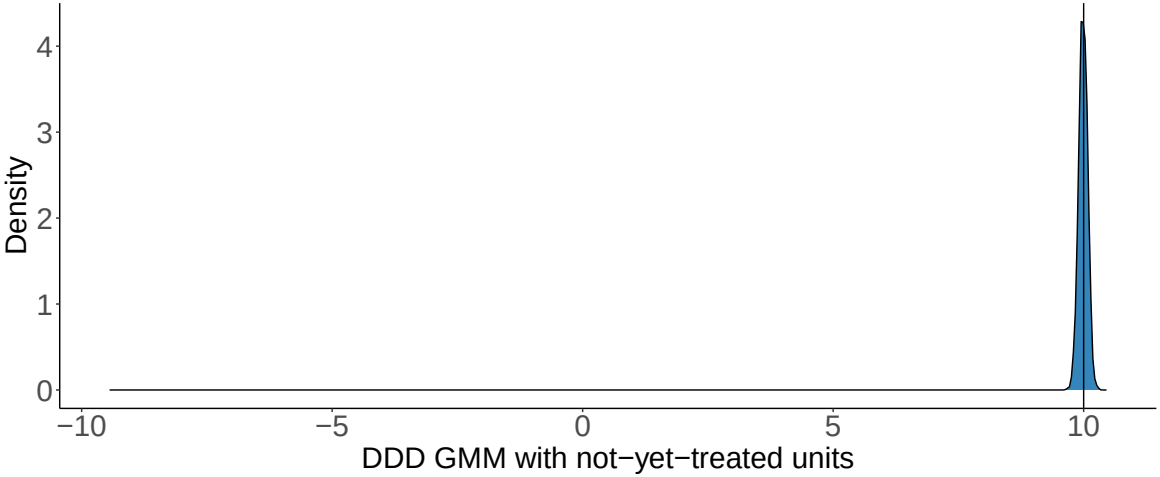
Challenging folks' wisdom: DDD using pooled not-yet-treated units



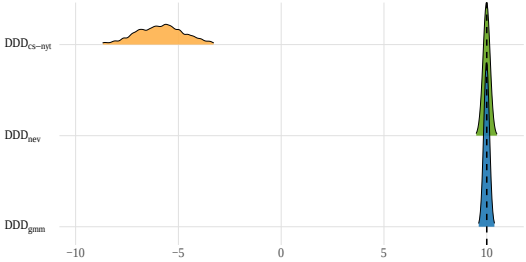
Challenging folks' wisdom: DDD using never-treated units



Challenging folks' wisdom: DDD GMM using all not-yet-treated units



Challenging folks' wisdom: DR DDD with staggered adoption!



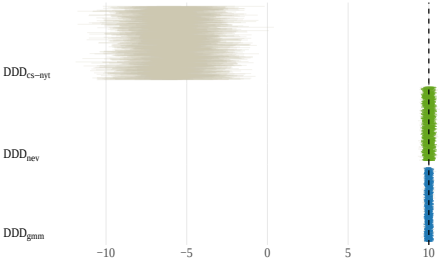


Table 2: Comparison of estimators across metrics

Estimator	Relative Bias	Relative RMSE	Coverage	Relative ALCI
DDD_{gmm}	1.00	1.00	0.95	1.00
DDD_{nev}	1.33	1.57	0.93	1.51
$DDD_{\text{cs-nyt}}$	1808.88	185.38	0.00	14.46

The Econometrics of DDD

Notation: panel and treatment timing

We observe a panel of n units indexed by $i = 1, \ldots, n$ over $t = 1, \ldots, T$ periods.

- Each unit may be exposed to a *binary*, *absorbing* treatment in any period $t > 1$.
- **Treatment enabling**: $S_i \in \mathcal{S} \subset \{2, \ldots, T\} \cup \{\infty\}$ denotes the first period the policy is enabled for unit i's group.
- **Eligibility**: $Q_i = 1$ if unit i qualifies within its group, $Q_i = 0$ otherwise.
- **Treatment indicator**: $D_{it} = 1\{t \geq S_i,\ Q_i = 1\}$, with *first treatment period* $G_i = \min\{t : D_{it} = 1\}$.

  Example: units with $S_i = 2$ and $Q_i = 1$ are treated from $t = 2$ onward ($G_i = 2$); otherwise they remain untreated ($G_i = \infty$).
- **Covariates**: pre-treatment vector $X_i \in \mathcal{X} \subseteq \mathbb{R}^d$.

Potential Outcomes + Parameters of Interest

- Let $\mathbf{0}_s$ and $\mathbf{1}_s$ be s-dimensional vectors of zeros and ones, respectively.
- Let $Y_{i,t}(\mathbf{0}_{g-1}, \mathbf{1}_{T-g+1})$ denote the potential outcome for unit i at time t if unit i is first treated at time g; and $Y_{i,t}(\mathbf{0}_T)$ the outcome if untreated by time $t = T$.
- To ease the notation, $Y_{i,t}(g) = Y_{i,t}(\mathbf{0}_{g-1}, \mathbf{1}_{T-g+1})$ and $Y_{i,t}(\infty) = Y_{i,t}(\mathbf{0}_T)$. We observe $Y_{i,t} = \sum_{g \in \mathcal{G}} 1\{G_i = g\} Y_{i,t}(g)$

Definition (Group-time Average Treatment Effect on the Treated)

$$ATT(g,t) \equiv \mathbb{E}[Y_{i,t}(g) - Y_{i,t}(\infty) | G_i = g] = \mathbb{E}[Y_{i,t}(g) - Y_{i,t}(\infty) | S_i = g, Q_i = 1]$$

- **Identification Problem:** We don't observe $\mathbb{E}\left[Y_{i,t}(\infty) | S_i = g, Q_i = 1\right]$ in $t \geq g$.

Definition (Aggregated Event Study Parameter)

$$ES(e) \equiv \mathbb{E}\left[ATT(G, G+e) \,\middle|\, G + e \in [2, T]\right] = \sum_{g \in \mathcal{G}_{\text{trt}}} \mathbb{P}(G = g | G + e \in [2, T]) \cdot ATT(g, g+e)$$

Assumptions

- Adding covariates linearly to 3WFE imposes two hidden restrictions: *homogeneous* covariate effects and no *covariate-specific trends*.
- Instead, we impose a *conditional* DDD PTA-type which is valid after controlling for covariates X:

Assumption (DDD-Conditional Parallel Trends)

For each $g \in \mathcal{G}_{trt}$, $g' \in \mathcal{S}$ and time periods t such that $t \geq g$ and $g' > \max\{g, t\}$, with probability one,

$$\begin{aligned} \mathbb{E}\left[Y_t(\infty) - Y_{t-1}(\infty) | S = g, Q = 1, X\right] &- \mathbb{E}\left[Y_t(\infty) - Y_{t-1}(\infty) | S = g, Q = 0, X\right] \\ &= \\ \mathbb{E}\left[Y_t(\infty) - Y_{t-1}(\infty) | S = g', Q = 1, X\right] &- \mathbb{E}\left[Y_t(\infty) - Y_{t-1}(\infty) | S = g', Q = 0, X\right]. \end{aligned}$$

Additional Assumptions: **A.2** Random Sampling **A.3** No Anticipation **A.4** Strong Overlap

View Assumptions

Efficiency and Double Robustness

- Under the previous assumptions, the $ATT(2,2)$ can be identified via *regression adjustments* or *IPW* or any convex combination between them. RA & IPW

- However, this depends on the researcher's ability to accurately model outcome regression or propensity scores.
  - How would you choose this combination if the goal was to achieve **doubly robustness** (DR)?
  - How would you choose this combination if the goal was **efficiency**?

- We tackle these questions by deriving the *efficient influence function* (EIF) for the ATT in DDD setups. EIF Derivation

- That usually leads to DR estimands, too.

DR DDD as a function of 3 DiDs

- By simple manipulation of the EIF, we get the following *DR-DDD* estimand for the $ATT(2,2)$,

$$
\begin{aligned}
ATT_{\text{dr}}(2,2) = \; & \mathbb{E}\left[\left(w_{\text{trt}}^{S=2,Q=1} - w_{\text{comp}}^{S=2,Q=0}(X)\right)\left(Y_2 - Y_1 - m_{Y_2-Y_1}^{S=2,Q=0}(X)\right)\right] \\
& + \mathbb{E}\left[\left(w_{\text{trt}}^{S=2,Q=1} - w_{\text{comp}}^{S=\infty,Q=1}(X)\right)\left(Y_2 - Y_1 - m_{Y_2-Y_1}^{S=\infty,Q=1}(X)\right)\right] \\
& - \mathbb{E}\left[\left(w_{\text{trt}}^{S=2,Q=1} - w_{\text{comp}}^{S=\infty,Q=0}(X)\right)\left(Y_2 - Y_1 - m_{Y_2-Y_1}^{S=\infty,Q=0}(X)\right)\right]
\end{aligned}
$$

- For estimation, we follow a two-step approach:
  - (1) estimate the nuisance functions $p(\cdot)$ and $m(\cdot)$,
  - (2) plug-in into the sample counterpart of the previous estimand to get an estimator for $ATT(2,2)$.

What about for *multiple time periods* and *variation in treatment timing*?

DR DDD estimator for $ATT(g,t)$ with any g_c comparison group

- Let our nuisance functions be

$$m^{S=g,Q=q}_{Y_t-Y_{t'}}(X) := \mathbb{E}\left[Y_t - Y_{t'} | S=g, Q=q, X\right];$$
$$p^{S=g,Q=1}_{g',q'}(X) := \mathbb{P}[S=g, Q=1 | X, (S=g, Q=1) \cup (S=g', Q=q')].$$

- For any $g_c \in \mathcal{S}$ such that $g_c > \max\{g,t\}$, and any *post-treatment* period $t \geq g$, let the DR DDD estimand for the $ATT(g,t)$ be given by

$$\begin{aligned} ATT_{\mathrm{dr},g_c}(g,t) = &\mathbb{E}\left[\left(w^{S=g,Q=1}_{\mathrm{trt}}(S,Q) - w^{S=g,Q=1}_{g,0}(S,Q,X)\right)\left(Y_t - Y_{g-1} - m^{S=g,Q=0}_{Y_t-Y_{g-1}}(X)\right)\right] \\ &+ \mathbb{E}\left[\left(w^{S=g,Q=1}_{\mathrm{trt}}(S,Q) - w^{S=g,Q=1}_{g_c,1}(S,Q,X)\right)\left(Y_t - Y_{g-1} - m^{S=g_c,Q=1}_{Y_t-Y_{g-1}}(X)\right)\right] \\ &- \mathbb{E}\left[\left(w^{S=g,Q=1}_{\mathrm{trt}}(S,Q) - w^{S=g,Q=1}_{g_c,0}(S,Q,X)\right)\left(Y_t - Y_{g-1} - m^{S=g_c,Q=0}_{Y_t-Y_{g-1}}(X)\right)\right] \end{aligned}$$

Weights notation and additional estimands

Over-identification and Optimal GMM-based DDD

- The model is *over-identified*: any weighted average across valid g_c's identifies $ATT(g,t)$ (Chen et al., 2025). Overidentification Results

  Example: To identify $ATT(2,2)$ when $S \in \{2,3,\infty\}$, we can use either $g_c = 3$ or $g_c = \infty$ as the comparison group; both yield the same valid result under our assumptions.
  - **However, Pooling all not-yet-treated would not work!**

- We exploit over-identification via GMM to obtain the *minimum variance* DDD estimator.

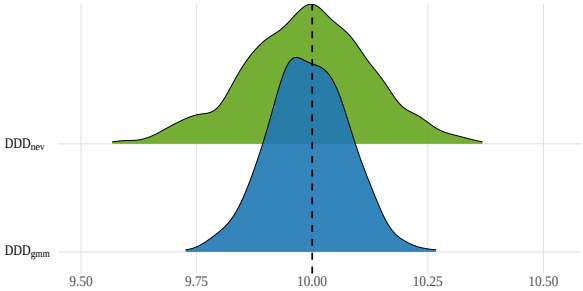
From standard GMM (Newey and McFadden, 1994), the optimal (population) GMM estimator is:

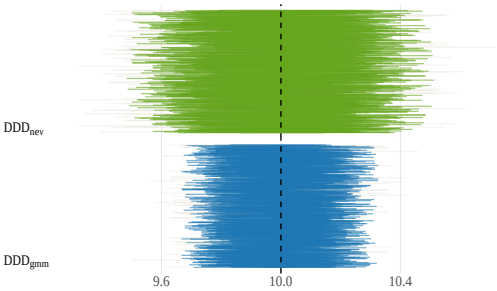
$$ATT_{\text{dr,opt}}(g,t) = \frac{\mathbf{1}'\Omega_{g,t}^{-1}}{\mathbf{1}'\Omega_{g,t}^{-1}\mathbf{1}} ATT_{\text{dr}}(g,t),$$

where $\Omega_{g,t} \equiv \mathbb{E}[\mathbb{IF}_{\text{dr}}(g,t) \cdot \mathbb{IF}_{\text{dr}}(g,t)']$ is positive definite and $\mathbb{IF}_{\text{dr}}(g,t)$ stacks the influence functions over all valid $g_c > \max\{g,t\}$.

How these *efficiency gains* look like?

Efficiency Gains





Asymptotic theory for $ATT(g,t)'s$

Key takeaways

- DDD is widely used in empirical research, but its properties have received little attention.
- In its simplest form, it is equivalent to estimating two separate DiD estimators and subtracting one from the other.
  - This equivalence breaks down when covariates play an important role in the analysis.
- We use semiparametric efficiency theory to forward-engineering DR DDD estimands and circumvent the issues pointed out.
  - We can even use ML algorithms to estimate the nuisances parameters!
- We can leverage these results to tackle staggered treatment setups, too.
  - However, we need to be clever on how we use the information available.

Scan Me

- For more details, the WP is already available on ArXiv.
- Our `tripledi ff` package that automates all these procedures is also available on CRAN for practitioners!

Working paper

R package

Heterogeneous Moderating Effects in Difference-in-Differences with Continuous Treatment

Heterogeneity analysis: Why it matters?

- Knowing *who* responds to a treatment is as important as knowing *whether* it works
  - It informs economic mechanisms, guides targeting, and enables personalized policy.
- Despite recent progress in the DiD literature (Roth et al., 2023; Arkhangelsky and Imbens, 2024), *heterogeneity analysis* in these settings remains underdeveloped.
- Many policies are *continuous* in nature and exhibit heterogeneous effects even at identical treatment intensities, a dimension the existing literature largely ignores.

Research Question: What *identification assumptions* and *estimation strategies* allow researchers to recover how continuous treatment effects vary with a pre-specified *moderating covariate* without imposing functional form restrictions?

Example: Roberto et al. (2019)

- **Program**: Effect of Philadelphia's *1.5¢/oz beverage tax* (Jan 2017) on sugar-sweetened drink consumption.
- **Design**: Jan 2016–Dec 2017 panel comparing Philadelphia stores to Baltimore, MD (no tax).
- **Setting**:
  - *Treatment*: distance from store's ZIP centroid to nearest tax-exempt ZIP.
  - *Moderator*: pre-tax neighborhood socioeconomic index.

Fig 2: Philly Bev Tax (The Philadelphia Inquirer)

> ***Core Idea:*** Move beyond asking, "Does this dosage (treatment) work?" to asking,
> $\longrightarrow$ **"For whom does the dosage work?"**

Where does this paper fit? The gap in the literature

The current state of the modern DiD literature includes the following settings (in a non-exhaustive sense):

- Binary Treatment + Panel Data + No Heterogeneity
  - Most typical case. See Roth et al. (2023); Callaway (2023) for a review.
- Binary Treatment + Repeated Cross-sectional Data + No Heterogeneity
  - Abadie (2005); Sant'Anna and Zhao (2020); Callaway and Sant'Anna (2021); Sant'Anna and Xu (2026)
- Binary Treatment + Panel Data + **Heterogeneity**
  - Imai et al. (2025)
- Binary Treatment + Repeated Cross-sectional Data + **Heterogeneity**
  - Nie et al. (2021)
- **Continuous Treatment** + Panel Data + No Heterogeneity
  - Callaway et al. (2024); de Chaisemartin et al. (2024, 2025)
- **Continuous Treatment** + Repeated Cross-sectional Data + No Heterogeneity
  - D'Haultfœuille et al. (2023); Haddad et al. (2026)

How can we incorporate *heterogeneity analysis* into DiD with a *continuous treatment*?

How existing tools handle this and why they fall short

Standard practice (Baker et al., 2025): partition X_{mod} into bins, $\text{Bin}(X_{\text{mod}}) \in \{1, 2, \ldots P\}$, and run

1. Interacted TWFE: $Y_{i,t} = \alpha_i + \gamma_t + \sum_{p=1}^{P} \beta_p \cdot D_i \cdot Post_t \cdot 1\{\text{Bin}(X_{\text{mod}}) = p\} + X'_{i,t}\delta + \varepsilon_{i,t}$
2. Split-sample: $Y_{i,t} = \alpha_i^{(p)} + \gamma_t^{(p)} + \beta_p \cdot D_i \cdot Post_t + X'_{i,t}\delta^{(p)} + \varepsilon_{i,t}^{(p)}$

- **Limitation 1**: Bin boundaries may be *arbitrary* $\longrightarrow$ Ex: which age bins for $X_{\text{mod}} = \text{AGE}$?
- **Limitation 2**: (1) and (2) are *not equivalent* $\longrightarrow$ global γ_t vs. partition-specific $\gamma_t^{(p)}$ imposes different trend restrictions.
- **Limitation 3**: β_p may have undesirable properties as target parameter and may require stronger assumptions for causal interpretation (Callaway et al., 2024).
- **Limitation 4**: Neither accommodates a *continuous* moderator with a *continuous* treatment without additional parametric assumptions.

Setup

For simplicity, consider a 2×2 DiD with n units and 2 time periods of *panel data*: $t \in \{1, 2\}$.

- **Treatment**: in $t = 2$, unit i receives dosage $D_i \in \mathcal{D} \subseteq \{0\} \cup (0, \infty)$ with $\mathbb{P}(D = 0) > 0$; no one is treated in $t = 1$.
- **Pre-treatment Covariates**: $X_i = [X_{\text{mod}}, X_{-\text{mod}}] \in \mathcal{X} \subset \mathbb{R}^k$, where X_{mod} is the *moderator* of interest and $X_{-\text{mod}}$ adjusts for confounding (i.e., PTA holds after conditioning on covariates).
- **Potential outcomes**: $Y_{i,t}(d)$ denotes unit i's outcome at time t under dose d, with $Y_{i,t}(0)$ the untreated potential outcome. We observe $Y_{i,t} = Y_{i,t}(D_i)$.

Parameters of Interest and Identification Result

Definition (Conditional Marginal Dose Response Function)

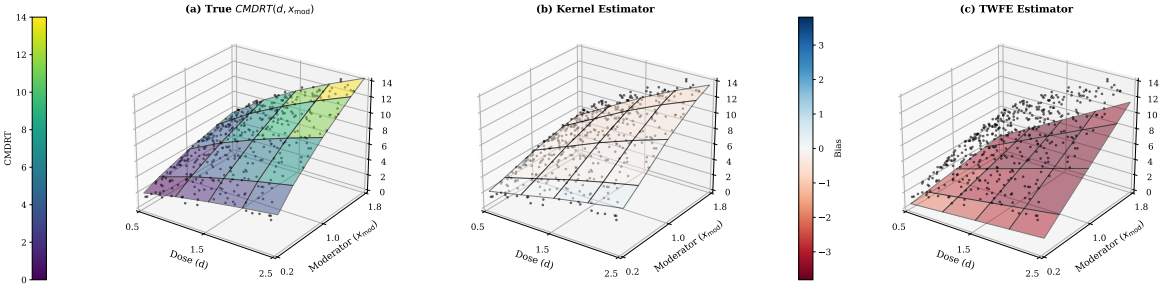
$$\text{CMDRT}(d, x_{\text{mod}}) \equiv \mathbb{E}\left[Y_2(d) - Y_2(0) \mid D = d, X_{\text{mod}} = x_{\text{mod}}\right], \quad (d, x_{\text{mod}}) \in \mathcal{D} \times \mathcal{X}_{\text{mod}}.$$

- Under conditional PTA for dose d, no anticipation, and strict overlap, $\text{CMDRT}(d, x_{\text{mod}})$ is identified for any $d \in \mathcal{D}_+$ and $x_{\text{mod}} \in \mathcal{X}_{\text{mod}}$ by: See Assumptions

$$\mathbb{E}[\Delta Y \mid D = d, X_{\text{mod}} = x_{\text{mod}}] - \mathbb{E}\left[\Delta Y \cdot 1\{D = 0\} \cdot \frac{f_{D|X}(d \mid X)}{f_{D|X_{\text{mod}}}(d \mid x_{\text{mod}})\mathbb{P}(D = 0 \mid X)} \,\middle|\, X_{\text{mod}} = x_{\text{mod}}\right]$$

  where $\Delta Y = Y_2 - Y_1$.
- This result extends the density reweighting argument to the joint distribution conditioning on $(d, x_{\text{mod}}) \in \mathcal{D}_+ \times \mathcal{X}_{\text{mod}}$.

Proof-of-Concept: When naive extrapolation may fail



Simulation Design

The danger of extrapolating existing tools

- The simulation shows that TWFE *severely misspecified* the response surface when treatment effects vary heterogeneously across dose and moderator.
- Even if practitioners only report bin-specific estimates and the full surface is not the target, the choice of bins remains arbitrary.
  - Different cutoffs can yield qualitatively different conclusions about *who* benefits from the policy.
- This motivates going beyond ex-ante binning and linear extrapolation entirely.

> **Goal**: How can we recover heterogeneous effects across moderator-defined subgroups without pre-specifying bins or functional forms?

Next Steps

- Estimating $\mathrm{CMDRT}(d, x_{\mathrm{mod}})$ nonparametrically is data-intensive and subject to the curse of dimensionality when both d and x_{mod} are continuous.
- Additionally, analyzing *response surfaces* of functionals can become more complicated when dealing with irregular geometries.
- Potential solutions: learn data-driven $\mathrm{Bin}(\cdot)$ maps without pre-specifying the number of subgroups ex-ante, via:
  - Clustering à la Bonhomme and Manresa (2015) to capture grouped heterogeneity patterns.
  - Tree-based methods à la Athey et al. (2019) to partition the moderator space adaptively.
- Either approach yields cluster-specific curves $\mathrm{CMDRT}(d, x^{c}_{\mathrm{mod}})$, which will be easier to analyze graphically.

Flexible Reweighting and Matrix Completion Methods in Nonlinear Causal Panel Data

Moving away from the PTA

- Standard DiD methods rely on a PTA and employ TWFE specifications as the primary empirical workhorse.
- However, TWFE involves rigid functional forms and is not robust to treatment effect heterogeneity, so we may want to allow for richer *time patterns* and *unit-specific* dynamics.
- One possible generalization is to model outcomes using *latent factor models*

Latent factor models posit the following family of outcome models for the outcome process

$$Y_{i,t} = h(F_{it}) + f(X_{it}) + \varepsilon_{it}.$$

where $h(\cdot)$ and $f(\cdot)$ are known, parametric functions. It encompasses several panel data models:

- **TWFE**: $h(F_{it}) = \alpha_i + \gamma_t$ and $f(X_{i,t}) = X'_{i,t}\beta$.
- **Interacted Fixed Effect** (IFE): $h(F_{i,t}) = \Lambda'_i \Gamma_t$ and $f(X_{i,t}) = X'_{it}\beta$, where $\Gamma_t = [f_{1t}, \cdots, f_{rt}]'$, $\Lambda_i = [\lambda_{i1}, \cdots, \lambda_{ir}]'$ and r is a fixed rank.

Quick overview: low-rank structure on untreated potential outcomes

- SOTA panel data estimators for causal inference like *SDID*, *MC*, and *TROP* are based on the latent factor model described previously[2]. Overview of these methods

- Such estimators assume the *untreated potential outcome* can be modeled based on the following linear structure:

$$Y_{i,t}(0) = \underbrace{\mathbf{L}_{i,t}}_{\Lambda_i'\Gamma_t} + X_{i,t}\beta + \varepsilon, \qquad \mathbb{E}[\varepsilon_{i,t}|\mathbf{L}, \mathbf{X}] = 0.$$

> ***Core Idea:*** Impute counterfactual outcomes for treated units based on the behavior of control units that share similar sensitivities to common shocks.

[2] See Arkhangelsky et al. (2021a); Athey et al. (2021, 2025) for a complete discussion.

How can we leverage the *information available* to allow more *general outcomes models*?

Heterogeneity and nonlinearity beyond linear factor models

- Recent work shows that *nonlinear* factor structures are empirically important and can overturn conclusions from standard TWFE/IFE models.[3]
- Yet existing methods discard potentially valuable information from *covariates* and *pre-treatment outcomes* that could sharpen counterfactual estimation:
  - covariates enter the outcome model *linearly and additively*;
  - the factor structure is *independent* of covariates; and
  - treatment assignment mechanism is *independent* to observed characteristics.

Research Question: When unobserved confounders follow a *nonparametric, nonlinear, and non-separable* factor model, what identification conditions and estimation strategies recover average treatment effects with large panels?

[3] See e.g., Hsiang (2016), Bansal et al. (1993), Bobeica et al. (2025), Voigtländer (2014).

Setup and Parameter of Interest

We observe $N = N_{tr} + N_{co}$ units $i \in [N] := \{1, \ldots, N\}$ over $t \in [T] := \{1, \ldots, T\}$ periods, with treatment starting at $T_{pre} + 1$. Consider a panel with large T and N.

- **Group and treatment indicators**: $G_i = 1$ if treated, $G_i = 0$ otherwise; $D_{i,t} = 1\{t \geq T_{pre} + 1,\ G_i = 1\}$.
- **Potential outcomes**: $\{Y_{i,t}(1), Y_{i,t}(0)\}$ for unit i at time t.
- **Covariates**: A real vector $X_{i,t} \in \mathcal{X} \subseteq \mathbb{R}^d$, collected for the unit-time pair (i, t).

Definition (Average Treatment Effect on the Treated)

$$ATT(t) = \mathbb{E}[Y_{i,t}(1) - Y_{i,t}(0) \mid G_i = 1], \quad T_{pre} < t \leq T.$$

- **Identification Problem**: $Y_{i,t}(0)$ is unobserved for treated units, so we are aiming to impute these *missing not at random values*. Overview of MNAR matrix completion

Model

- Let untreated potential outcomes follows a possibly nonlinear, non-separable factor model[4]

$$Y_{i,t}(0) = g(\lambda_i, f_t, X_{it}) + \varepsilon_{i,t}, \quad \mathbb{E}[\varepsilon_{i,t}|\lambda_i, f_t, X_{i,t}].$$

  where λ_i and f_t are *unobserved* unit and time FE and $g(\cdot)$ is an *unknown* function.

- Such outcome models include TWFE and IFE models as special cases.
- **Key challenge:** without restricting $g(\cdot)$, unit heterogeneity λ_i and common factors f_t are not separately identified from the nonlinear interaction $g(\lambda_i, f_t, X_{it})$.

[4] See, e.g., Hoderlein and White (2012); Fernández-Val et al. (2021); Bonhomme et al. (2022); Ditzen and Karavias (2025); Maschmann and Westerlund (2025); Feng (2021)

But...what would happen with SOTA models if we move away from a *linear factor model*?

Proof-of-concept: Simulations

- Large panel with $N = 50$, $T_{pre} = 50$, and one post-treatment period. Let $\lambda_i, f_t \sim \text{Unif}(-1, 1)$, $\varepsilon_{i,t} \sim \text{N}(0, 0.25)$, and true effect $\tau = 0.5$.

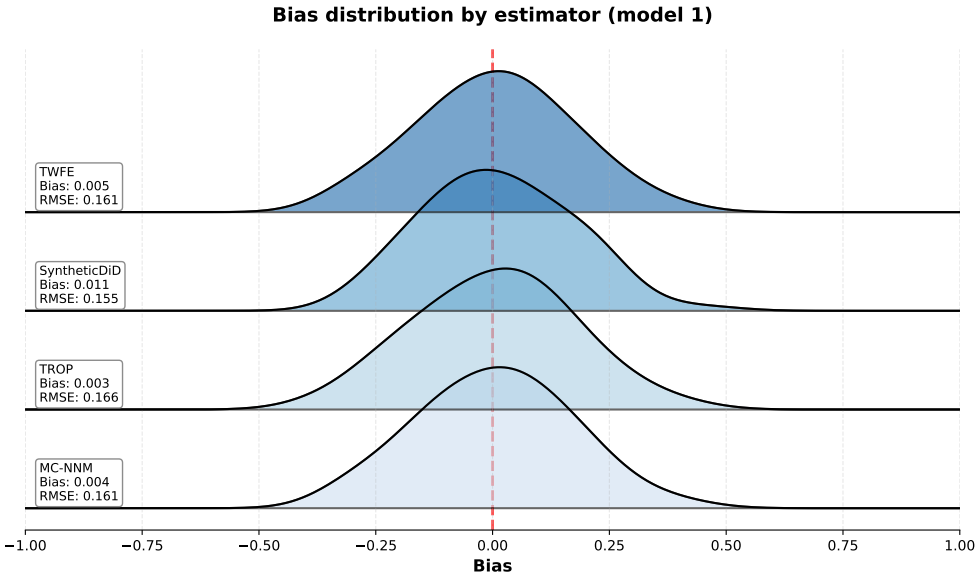
We consider three data-generating processes for $Y_{i,t}(0)$:

1. **TWFE**: $Y_{i,t}(0) = \lambda_i + f_t + \varepsilon_{i,t}$
2. **IFE**: $Y_{i,t}(0) = \lambda_i \cdot f_t + \varepsilon_{i,t}$
3. **Nonlinear**: $Y_{i,t}(0) = \begin{cases} (\lambda_i - f_t)^2 + \varepsilon_{i,t} & t \leq T_{pre} \\ \lambda_i(\lambda_i + 1) + \varepsilon_{i,t} & t = T \end{cases}, \quad Y_{i,T}(1) = (\lambda_i + 1)^2 + \varepsilon_{i,t}$

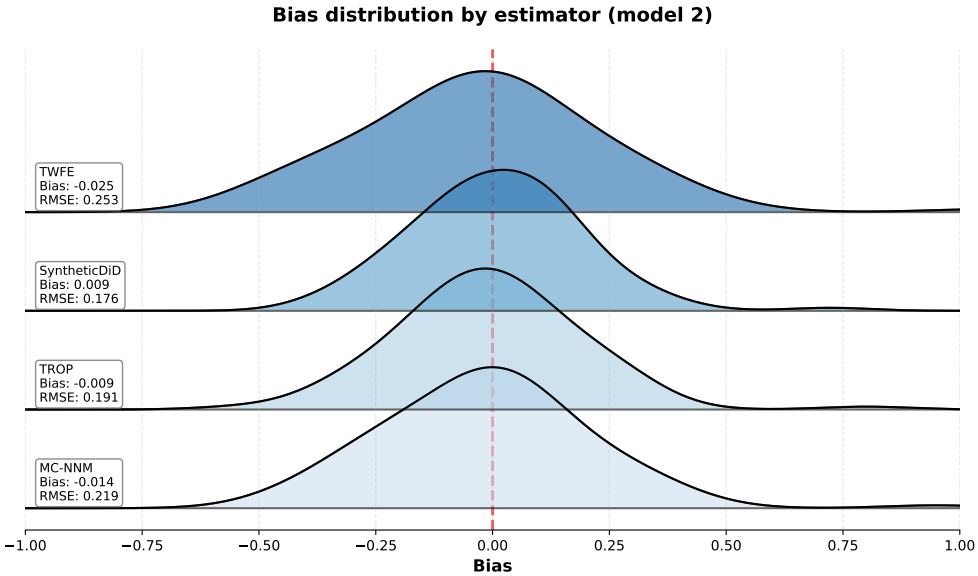
- Treatment Assignment: $G_i | \lambda_i, \varepsilon_{i,t} \sim \text{Bernoulli}(\pi_i)$, where $\pi_i = \text{expit}(\phi_\lambda \lambda_i)$.

Note: These DGPs are adaptations from the ones in Arkhangelsky et al. (2021b); Fernández-Val et al. (2021); Feng (2021).

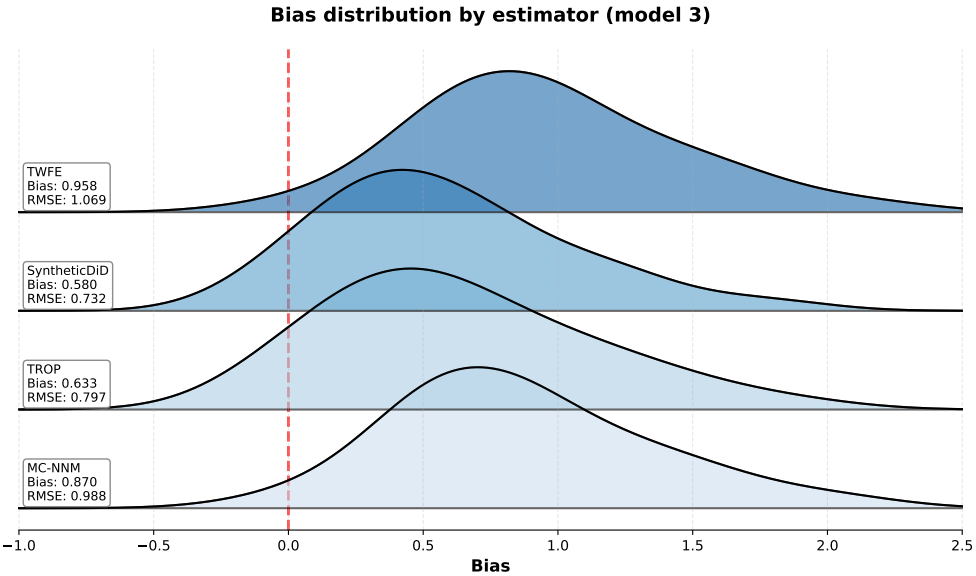
Outcome model: TWFE



Outcome model: IFE



Outcome model: Nonlinear



Proposal: low-rank based vs. neighbor-based

- There are two major strands of literature on imputation methods for matrix completion: low-rank based and neighbor-based[5].
- This paper follows a *neighbor-based* approach suitable for nonlinear factor models.
- One could try to naively estimate the function $g(\cdot)$ with nonparametric methods by kernel smoothing around similar λ_i. Ex:

$$\hat{g}(\lambda_i, \cdot, \cdot) = \frac{\sum K_h(\lambda_i - \lambda_j) Y_{i,t}}{\sum K_h(\lambda_i - \lambda_j)}$$

- The problem? We don't observe λ_i.

[5] Recent papers have proposed imposing smooth requirements to $g(\cdot)$, see e.g., Agarwal et al. (2021); Fernández-Val et al. (2021); Feng (2021); Athey and Imbens (2025)

Proposal: The Core Idea

- Assume pre-treatment outcomes $Y_{i,1:T_{pre}}$ and pre-treatment covariates $X_{i,1:T_{pre}}$ carry information about λ_i
- Units with similar pre-treatment trajectories likely share similar factor loadings.

Key insight: replace smoothing over λ_i with smoothing over pre-treatment histories, using a neighborhood smoothing distance on $(Y_{i,1:T_{pre}}, X_{i,1:T_{pre}})$.

Proposal: Neighborhood Smoothing

- Feng (2021) proposed an imputation method by leveraging a *neighborhood smoothing* distance (Zhang et al., 2017) to find K nearest neighbors of unit i

$$\mathcal{N}_i = \left\{ j_k(i) : \sum_{j=1}^{n} 1\left(\rho\left(Y_i, Y_j\right) \leq \rho\left(Y_i, Y_{j_k(i)}\right)\right) \leq K, 1 \leq k \leq K \right\}.$$

- It uses the $K-$neighbors in $\mathcal{N}_i$ for estimating the factors loadings with kernel PCA and then perform the counterfactual imputation.

- Here, I'm aiming to impute the counterfactual directly without estimating factors.

> ***How?*** Use a flexible imputation methods for $\mathbf{Y}_{i,t}(0)$ over reproducing kernel Hilbert spaces in order to account for prior information present in the available data.

Proposal: Direct Imputation via Kernel Regression + Neighborhood Smoothing

- Following Giménez-Febrer et al. (2018), we postulate that row vectors of $\mathbf{Y}(0)$ belong to an RKHS, where kernel functions encode similarity between units.
- RKHSs allow flexible, nonparametric approximation of $g(\cdot)$ while remaining computationally tractable
- The kernel is built from a *neighborhood smoothing* distance on pre-treatment histories:

Let $W_{i,t} = (Y_{i,t}, X_{i,t})$. Using the information of pre-treatment periods, we compute the distance of two i, i' units as

$$\rho(i, i') = \max_{k \notin \{i,j\}} \left| \left\langle W_{k,1:T_{pre}}, W_{i,1:T_{pre}} - W_{i',1:T_{pre}} \right\rangle \right|$$

Example

- **Claim:** Units close under $\rho(\cdot, \cdot)$ share similar latent dynamics, making the imputation feasible without ever estimating λ_i or f_t.

Thanks!

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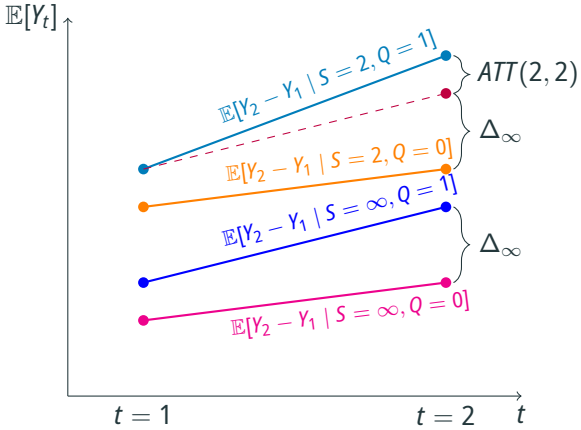
Appendix

$2x2x2$ DDD without covariates

Let, $g \in S = \{2, \infty\}$ and $Q = \{0, 1\}$.



$$
\begin{aligned}
\theta = \Big[& \Big(\mathbb{E}[Y_2 | S = 2, Q = 1] - \mathbb{E}[Y_1 | S = 2, Q = 1] \Big) \\
& - \Big(\mathbb{E}[Y_2 | S = 2, Q = 0] - \mathbb{E}[Y_1 | S = 2, Q = 0] \Big) \Big] \\
& - \Big[\Big(\mathbb{E}[Y_2 | S = \infty, Q = 1] - \mathbb{E}[Y_1 | S = \infty, Q = 1] \Big) \\
& - \Big(\mathbb{E}[Y_2 | S = \infty, Q = 0] - \mathbb{E}[Y_1 | S = \infty, Q = 0] \Big) \Big]
\end{aligned}
$$

- Note that this is the difference of two DiD's: one among $S = 2$ across Q groups, and one among $S = \infty$ across Q groups.

Go Back

WGAN-Based Data Generation

- **Outcome** (Y_{it}): Log wages.
- **Covariates** (X_i): nonwhite, union, education, experience, and industry dummies (13 categories).
- **Data Generator**: Wasserstein GANs learn the *marginal distribution of covariates* X and the *outcome distribution* $Y|S, Q, T, X$ by training a Generator to produce artificial data that mimics a real dataset.
  - It does this by minimizing the Wasserstein distance between the real and generated data distributions in a "min-max" game (Athey et al., 2024):

$$\min_{\theta_g} \max_{\theta_c} \left(\underbrace{\frac{1}{N_R} \sum_{i=1}^{N_R} f\left(X_i; \theta_c\right)}_{\text{Real Score}} - \underbrace{\frac{1}{N_F} \sum_{i=1}^{N_F} f\left(g\left(Z_i; \theta_g\right); \theta_c\right)}_{\text{Fake Score}} \right)$$

- Let $g_X(Z_x; \theta_x^*)$ and $g_Y(Z_y; \theta_{y|s,q,t,x}^*)$ be the learned generator functions for covariates and outcomes for a specific group-time combination such that:

$$\tilde{X} = g_X\left(Z_x; \theta_x^*\right) \sim \mathbb{P}_n(X); \quad \tilde{Y}_{s,q,t} = g_Y\left(Z_y; \theta_{y|s,q,t,x}^*\right) \sim \mathbb{P}_n(Y \mid S = s, Q = q, T = t, X)$$

Monte Carlo Design: Data Generation Process

For every simulated observation i, the following steps are executed sequentially:

- **Step 1. Generate Covariates:** Draw a latent noise vector $Z_{i,x}$ and generate the covariate vector using the first WGAN:

$$\tilde{X}_i = g_X\left(Z_{i,x}; \theta_x^*\right)$$

- **Step 2. Assign Group Membership** (S, Q)**:** Estimate a Multinomial Logit model on the original data to model the probability of belonging to the four partitions defined by S and Q. With 4 partitions, the conditional probability that each unit belongs to subgroup $(g, q) \in \{2, \infty\} \times \{0, 1\}$ is:

$$p^{S=g,Q=g}(\tilde{X}_i) \equiv \mathbb{P}(S_i = g, Q_i = q \mid \tilde{X}_i) := \frac{\exp\left(\tilde{X}_i' \hat{\beta}_{g,q}\right)}{\sum_{(g,q)\in\{2,\infty\}\times\{0,1\}} \exp\left(\tilde{X}_i' \hat{\beta}_{g,q}\right)},$$

  where $\hat{\beta}$ are the empirical-calibrated coefficients from CPS data and heterogeneous across sub-groups.

Monte Carlo Design: Data Generation Process

Let $U \sim$ Uniform $[0,1]$, independent of all other variables. The partition groups are assigned as follows

$$(\tilde{S}, \tilde{Q}) := \begin{cases} (\infty, 0), & \text{if } U \leqslant p^{S=\infty,Q=0}(\tilde{X}), \\ (\infty, 1), & \text{if } p^{S=\infty,Q=0}(\tilde{X}) < U \leq \sum_{j=0}^{1} p^{S=\infty,Q=j}(\tilde{X}), \\ (2, 0), & \text{if } \sum_{j=0}^{1} p^{S=\infty,Q=j}(\tilde{X}) < U \leq \sum_{j=0}^{1} p^{S=\infty,Q=j}(\tilde{X}) + p^{S=2,Q=0}(\tilde{X}), \\ (2, 1), & \text{if } \sum_{j=0}^{1} p^{S=\infty,Q=j}(\tilde{X}) + p^{S=2,Q=0}(\tilde{X}) < U, \end{cases}$$

- **Step 3: Assign Time Period (T)**: Randomly assign the time period based on the marginal probability observed in the original data

$$\tilde{T}_i \sim \text{Bernoulli}\left(\mathbb{P}_n\left(T = 2\right)\right)$$

- **Step 4. Generate Observed Outcomes**: Conditioned on the generated $\tilde{X}, \tilde{S}, \tilde{Q}, \tilde{T}$, draw a noise vector $Z_{i,y}$ and generate a random draw for the outcome using the WGAN approximating $Y|S = g, Q = q, T = t, X$:

$$\tilde{Y}_{S=\tilde{g},Q=\tilde{q},T=\tilde{t}} = g_Y\left(Z_{i,y}; \theta^*_{y|s,q,t,x}\right)$$

Monte Carlo Design: Potential Outcomes

- To calculate an oracle ATT, we need the untreated potential outcome $Y_2(\infty)$ for the treated units $(S = 2, Q = 1)$.
- We explicitly construct this to satisfy the DDD-PT assumption.
- For a treated unit i (where $\tilde{S} = 2, \tilde{Q} = 1, \tilde{T} = 2$), the counterfactual $Y_2(\infty)$ is constructed via linear combination of draws from the different conditional WGANs as:

$$
\begin{aligned}
Y_{i,2}(\infty) = \tilde{Y}_{S=2,Q=1,T=1} \\
&+ \left(\tilde{Y}_{S=2,Q=0,T=2} - \tilde{Y}_{S=2,Q=0,T=1}\right) \\
&+ \left(\tilde{Y}_{S=\infty,Q=1,T=2} - \tilde{Y}_{S=\infty,Q=1,T=1}\right) \\
&- \left(\tilde{Y}_{S=\infty,Q=0,T=2} - \tilde{Y}_{S=\infty,Q=0,T=1}\right)
\end{aligned}
$$

- This imputation ensures that the generated counterfactual mechanically respects the DDD-PT assumption.

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Augmenting MC design to allow variation in treatment timing

- We also consider scenarios involving more than two periods.
- WLOG, let $t \in \{1, 2, 3\}$, three enabling groups, $S \in \{2, 3, \infty\}$, and two eligibility groups $Q \in \{0, 1\}$.
- To simplify, we exclude covariates in this context (don't panic, we address scenarios with covariates in the paper as well).
- For each unit i, the probabilities of belonging to each subgroup are: $p^{S=2,Q=0} = 0.20$, $p^{S=2,Q=1} = 0.15$, $p^{S=3,Q=0} = 0.30$, $p^{S=3,Q=1} = 0.20$, $p^{S=\infty,Q=0} = 0.05$, and $p^{S=\infty,Q=1} = 0.10$.
  - We then assign the realized value of (S, Q) randomly based on this distribution.
- We define the actual treatment effects as follows: $ATT(2, 2) = 10$, $ATT(2, 3) = 20$, and $ATT(3, 3) = 25$.

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Additional Assumptions

Assumption (Random Sampling)

$\{(Y_{i,t=1}, \ldots, Y_{i,t=T}, X_i', G_i, S_i, Q_i)'\}_{i=1}^n$ is a random sample from $(Y_{t=1}, \ldots, Y_{t=T}, X', G, S, Q)'$.

Assumption (No Anticipation)

For every $g \in \mathcal{G}_{trt}$, and every pre-treatment period $t < g$,
$\mathbb{E}[Y_{i,t}(g)|S = g, Q = 1, X] = \mathbb{E}[Y_{i,t}(\infty)|S = g, Q = 1, X]$ with probability one.

Assumption (Strong Overlap)

For every $(g, q) \in \mathcal{S} \times \{0, 1\}$ and for some $\epsilon > 0$, $\mathbb{P}[S = g, Q = q|X] > \epsilon$ with probability one.

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Regression Adjustment and IPW identification

We show in the paper that if conditional PT, no-anticipation, and strong overlap assumptions are satisfied and balanced panel data is available, the ATT is identified via *regression adjustments* or *IPW*:

$$ATT(2,2) = ATT_{ra}(2,2) = ATT_{ipw}(2,2),$$

where

$$ATT_{ra}(2,2) := \mathbb{E}\left[Y_2 - Y_1 | S = 2, Q = 1\right] - \mathbb{E}\left[m_{Y_2 - Y_1}^{S=2,Q=0}(X) + \left(m_{Y_2 - Y_1}^{S=\infty,Q=1}(X) - m_{Y_2 - Y_1}^{S=\infty,Q=0}(X)\right) \middle| S = 2, Q = 1\right],$$

$$ATT_{ipw}(2,2) := \mathbb{E}\left[\left(\left(w_{trt}^{S=2,Q=1} - w_{comp}^{S=2,Q=0}(X)\right) - \left(w_{comp}^{S=\infty,Q=1}(X) - w_{comp}^{S=\infty,Q=0}(X)\right)\right) \cdot (Y_2 - Y_1)\right].$$

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Efficient Influence Function

Proposition (Efficient Influence Function for DDD)

Suppose that conditional PT, no-anticipation, and strong overlap assumptions are satisfied, and balanced panel data is available. Let $\tau(\Delta Y, X) := \Delta Y - m^{S=2,Q=0}_{Y_2-Y_1}(X) - m^{S=\infty,Q=1}_{Y_2-Y_1}(X) + m^{S=\infty,Q=0}_{Y_2-Y_1}(X)$, $S := (\Delta Y, S, Q, X)$. *Then, the efficient influence function for the ATT*$(2,2)$ *is given by*

$$
\begin{aligned}
\eta_{eff}(S) = {} & w^{S=2,Q=1}_{trt} \cdot \Big(\tau(\Delta Y, X) - ATT(2,2)\Big) \\
& - w^{S=2,Q=0}_{comp}(X) \cdot \Big(\Delta Y - m^{S=2,Q=0}_{Y_2-Y_1}(X)\Big) \\
& - w^{S=\infty,Q=1}_{comp}(X) \cdot \Big(\Delta Y - m^{S=\infty,Q=1}_{Y_2-Y_1}(X)\Big) \\
& + w^{S=\infty,Q=0}_{comp}(X) \cdot \Big(\Delta Y - m^{S=\infty,Q=0}_{Y_2-Y_1}(X)\Big).
\end{aligned}
$$

Weights Notation Go back

Some additional notation

- For $(g, q) \in \{2, \infty\} \times \{0, 1\}$, let $\Delta Y = Y_2 - Y_1$, and

$$\begin{aligned} m_{Y_2-Y_1}^{S=g,Q=q}(X) &:= \mathbb{E}\left[Y_2 - Y_1 | S = g, Q = q, X\right], \quad \text{(outcome regression).} \\ p^{S=g,Q=q}(X) &:= \mathbb{P}[S = g, Q = q | X] \quad \text{(multi-valued propensity score).} \end{aligned}$$

- For $(g_c, q) \in \mathcal{S}_{comp} \equiv \{(\infty, 0), (\infty, 1), (2, 0)\}$, let

$$w_{trt}^{S=2,Q=1} := \frac{1_{\{S=2,Q=1\}}}{\mathbb{E}[1_{\{S=2,Q=1\}}]},$$

$$w_{comp}^{S=g_c,Q=q}(X) := \frac{\dfrac{1_{\{S=g_c,Q=q\}} \cdot p^{S=2,Q=1}(X)}{p^{S=g_c,Q=q}(X)}}{\mathbb{E}\left[\dfrac{1_{\{S=g_c,Q=q\}} \cdot p^{S=2,Q=1}(X)}{p^{S=g_c,Q=q}(X)}\right]}$$

- Let $\tau(\Delta Y, X) := \Delta Y - m_{Y_2-Y_1}^{S=2,Q=0}(X) - m_{Y_2-Y_1}^{S=\infty,Q=1}(X) + m_{Y_2-Y_1}^{S=\infty,Q=0}(X)$, $S := (\Delta Y, S, Q, X)$.

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Regression Adjustment and IPW identification for staggered DDD

Analogously, let the RA DDD estimand for the $ATT(g,t)$ be given by

$$ATT_{\text{ra},g_c}(g,t) = \mathbb{E}\left[w_{\text{trt}}^{S=g,Q=1}(S,Q)\left(Y_t - Y_{g-1} - m_{Y_t-Y_{g-1}}^{S=g,Q=0}(X) - m_{Y_t-Y_{g-1}}^{S=g_c,Q=1}(X) + m_{Y_t-Y_{g-1}}^{S=g_c,Q=0}(X)\right)\right]$$

and the IPW estimand be

$$\begin{aligned} ATT_{\text{ipw},g_c}(g,t) = &\mathbb{E}\left[\left(w_{\text{trt}}^{S=g,Q=1}(S,Q) - w_{g,0}^{S=g,Q=1}(S,Q,X)\right)(Y_t - Y_{g-1})\right] \\ &- \mathbb{E}\left[\left(w_{g_c,1}^{S=g,Q=1}(S,Q,X) - w_{g_c,0}^{S=g,Q=1}(S,Q,X)\right)(Y_t - Y_{g-1})\right]. \end{aligned}$$

where the weights w are given by

$$w_{\text{trt}}^{S=g,Q=1}(S,Q) := \frac{1\{S=g,Q=1\}}{\mathbb{E}[1\{S=g,Q=1\}]}, \quad w_{g',q'}^{S=g,Q=1}(S,Q,X) := \frac{\dfrac{1\{S=g',Q=q'\}\cdot p_{g',q'}^{S=g,Q=1}(X)}{1-p_{g',q'}^{S=g,Q=1}(X)}}{\mathbb{E}\left[\dfrac{1\{S=g',Q=q'\}\cdot p_{g',q'}^{S=g,Q=1}(X)}{1-p_{g',q'}^{S=g,Q=q}(X)}\right]}.$$

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Asymptotic theory for $ATT(g,t)'s$

Theorem (Consistency and Asymptotic Normality)

Let previous assumptions plus WM, ALR, and IC (see paper) hold. Then, for all $g \in \mathcal{G}_{trt}$, $t \in \{2,\ldots,T\}$, and $g_c \in \mathcal{G}_c^{g,t}$ such that $t \geq g$, provided that at least one working model is correctly specified,

$$\sqrt{n}\left(\widehat{ATT}_{dr,g_c}(g,t) - ATT(g,t)\right) = \frac{1}{\sqrt{n}}\sum_{i=1}^{n}\psi_{g_c}^{g,t}(W_i;\kappa_{0,g_c}^{g,t}) + o_p(1) \xrightarrow{d} N(0,\Omega_{g,t,g_c}),$$

where $\Omega_{g,t,g_c} = \mathbb{E}\left[\psi_{g_c}^{g,t}(W_i;\kappa_{0,g_c}^{g,t})\psi_{g_c}^{g,t}(W_i;\kappa_{0,g_c}^{g,t})'\right]$. Furthermore,

$$\sqrt{n}\left(\widehat{ATT}_{dr,opt}(g,t) - ATT(g,t)\right) = \frac{\mathbf{1}'\Omega_{g,t}^{-1}}{\mathbf{1}'\Omega_{g,t}^{-1}\mathbf{1}}\frac{1}{\sqrt{n}}\sum_{i=1}^{n}\psi^{g,t}(W_i;\kappa_0^{g,t}) + o_p(1) \xrightarrow{d} N(0,\Omega_{g,t,opt}),$$

where $\Omega_{g,t,opt} = \left(\mathbf{1}'\Omega_{g,t}^{-1}\mathbf{1}\right)^{-1} \leq \Omega_{g,t,g_c}$ for any $g_c \in \mathcal{G}_c^{g,t}$. In fact, for any set of weights w that sum up to one over the $\mathcal{G}_c^{g,t}$, $\Omega_{g,t,opt} \leq \Omega_{g,t,w}$, with $\Omega_{g,t,w}$ defined as the asymptotic variance of $\sqrt{n}\left(\widehat{ATT}_{dr,\widehat{w}}(g,t) - ATT_{dr,w}(g,t)\right)$.

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$ATT(g,t)$ is identified for all post-treatment periods

Theorem

Let previous assumptions hold. Then, for all $g \in \mathcal{G}_{trt}$, $t \in \{2,\ldots,T\}$, and $g_c \in \mathcal{S}$ such that $t \geq g$ and $g_c > t$,

$$ATT(g,t) = ATT_{dr,g_c}(g,t) = ATT_{ra,g_c}(g,t) = ATT_{ipw,g_c}(g,t).$$

Corollary

Let previous assumptions hold. Then, for all $g \in \mathcal{G}_{trt}$ and $t \in \{2,\ldots,T\}$ such that $t \geq g$, and any set of weights $w_{g_c}^{g,t}$ that sum up to one over $\mathcal{G}_c^{g,t}$,

$$ATT(g,t) = \sum_{g_c \in \mathcal{G}_c^{g,t}} w_{g_c}^{g,t} ATT_{dr,g_c}(g,t).$$

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Assumptions for $\text{CMDRT}(d, x_{\textbf{mod}})$

Assumption (Random Sampling)

The observed panel data consists of $\{Y_{i,t}, D_i, X_i : i = 1, \ldots, n; t = 1, 2\}$ *which is independent and identically distributed.*

Assumption (No Anticipation)

For every dosage $d \in \mathcal{D}$ *in the pre-treatment period,* $\mathbb{E}[Y_{i,t}(d)|X, D = d] = \mathbb{E}[Y_{i,t}(0)|X, D = d]$ *with probability one.*

Assumption (Strict Overlap for continuous D)

For all doses $d \in \mathcal{D}$ *and for some* $\epsilon > 0$, $f_{D|X}(d|x) > \epsilon$ *and* $\mathbb{P}(D = 0) > 0$.

Assumption (Conditional Parallel Trends)

For all $d \in \mathcal{D}_+ \subseteq (0, \infty)$, $\mathbb{E}\left[Y_2(0) - Y_1(0)|D = d, X = x\right] = \mathbb{E}\left[Y_2(0) - Y_1(0)|D = 0, X = x\right]$

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Quick simulation as proof of concept

- Covariates are generated as $X_{\text{mod}} \sim \mathit{Uniform}(0,2)$ and $X_{-\text{mod}} \sim N(0,1)$.
- Let $X = [X_{\text{mod}}, X_{-\text{mod}}]$ and $\alpha_i = 0.5X_{-\text{mod}} + 0.3X_{\text{mod}} + u_i, \quad u_i \sim N(0, 0.5^2)$
- **Treatment Selection** ($D_i > 0$ vs. $D_i = 0$): unit i is treated if

$$D_i^* = 1\{-0.3 + 0.4X_{-\text{mod}} + 0.3X_{\text{mod}} + v_i > 0\}, \quad v_i \sim N(0,1)$$

- **Dose**: $D_i \sim \begin{cases} \mathit{Exp}(\mu_i), & D_i^* = 1, \\ 0, & D_i^* = 0, \end{cases} \qquad \mu_i = \exp(0.3 + 0.2X_{-\text{mod}} + 0.15X_{\text{mod}})$

  (here $\mathit{Exp}(\mu_i)$ uses μ_i as the scale/mean parameter).
- **Potential Outcomes**: let $\varepsilon_{i1}, \varepsilon_{i2} \sim N(0, 0.5^2)$ and

$$\begin{aligned} Y_{i,1}(0) &= \alpha_i + 0.5X_{-\text{mod}} + 0.3X_{\text{mod}} + \varepsilon_{i1} \\ Y_{i,2}(0) &= 2 + \alpha_i + 0.7X_{-\text{mod}} + 0.5X_{\text{mod}} + \varepsilon_{i2} \\ Y_{i,2}(d) &= Y_{i,2}(0) + \text{CMDRT}(d, X_{\text{mod}}) \end{aligned}$$

- **Parameter of Interest**: $\text{CMDRT}(d, x_{\text{mod}}) = 6(d \cdot x_{\text{mod}})^{0.6}, \qquad \text{CMDRT}(0, x_{\text{mod}}) = 0$

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Quick Recap: SDID and MC

- SDID recognizes that outcomes from different periods can vary in how well they predict the target counterfactual.
- It uses nonnegative *unit weights* $\hat{\omega}^{\text{sdid}}$ and *time weights* $\hat{\theta}_t^{\text{sdid}}$ to "balanced out" the factors without estimating them as follows

$$\left(\widehat{\tau}^{\text{sdid}}, \widehat{\mu}, \widehat{\alpha}, \widehat{\beta}\right) = \underset{\tau,\mu,\alpha,\beta}{\arg\min} \left\{ \sum_{i=1}^{N} \sum_{t=1}^{T} (Y_{it} - \mu - \alpha_i - \beta_t - D_{it}\tau)^2 \, \widehat{\omega}_i^{\text{sdid}} \, \widehat{\theta}_t^{\text{sdid}} \right\}$$

- MC uses *nuclear norm regularization* to estimate the missing values of $\mathbf{Y}(0)$ matrix by shrinking the singular values of $\mathbf{L}$ solving the problem

$$\widehat{\mathbf{L}} = \underset{\mathbf{L}}{\arg\min} \left[\sum_{(i,t)\in\mathcal{O}} \frac{(Y_{it} - L_{it})^2}{|\mathcal{O}|} + \lambda_L \|\mathbf{L}\| \right]$$

where $\mathcal{O} = \{(i,t) \mid D_{it} = 0\}$, $\|\mathbf{L}\|$ is the chosen matrix norm of $\mathbf{L}$ and λ_L is a tuning parameter.

Unit & Time Weights | Missing Data Problem | Go back

How do we choose the weights $\hat{\omega}_i^{\text{sdid}}$ and $\hat{\theta}_t^{\text{sdid}}$?

- The optimization problem for the *unit weights* is given by

$$\left(\hat{\omega}_0, \hat{\omega}^{\text{sdid}}\right) = \underset{\omega_0 \in \mathbb{R}, \omega \in \Omega}{\arg\min} \sum_{t=1}^{T_{\text{pre}}} \left(\omega_0 + \sum_{i=1}^{N_{co}} \omega_i Y_{it} - \frac{1}{N_{\text{tr}}} \sum_{i=N_{co}+1}^{N} Y_{it} \right)^2 + \zeta^2 T_{\text{pre}} \|\omega\|_2^2 \tag{1}$$

where $\|\omega\|_2$ represents the Euclidean norm, ζ is a regularization parameter and

$$\Omega = \left\{ \omega \in \mathbb{R}_+^N, \text{ with } \sum_{i=1}^{N_{co}} \omega_i = 1 \text{ and } \omega_i = \frac{1}{N_{tr}} \text{ for all } i = N_{co}+1, \ldots, N \right\}.$$

- Arkhangelsky et al. (2021a) sets the regularization parameter as $\zeta = (N_{tr} \times T_{post})^{1/4} \hat{\sigma}$, where

$$\hat{\sigma}^2 = \frac{1}{N_{\text{co}} \left(T_{\text{pre}} - 1\right)} \sum_{i=1}^{N_{\text{co}}} \sum_{t=1}^{T_{\text{pre}}-1} \left(\Delta_{it} - \bar{\Delta}\right)^2,$$

$$\Delta_{it} = Y_{i,(t+1)} - Y_{it}, \text{ and } \bar{\Delta} = \frac{1}{N_{co} \left(T_{\text{pre}} - 1\right)} \sum_{i=1}^{N_{\text{co}}} \sum_{t=1}^{T_{\text{pre}}-1} \Delta_{it}.$$

How do we chose the weights $\hat{\omega}_i^{\text{sdid}}$ and $\hat{\theta}_t^{\text{sdid}}$?

- A similar optimization problem can be written for the time weights. For those weights, we want to minimize the following objective function

$$\left(\widehat{\theta}_0, \widehat{\theta}^{sdid}\right) = \underset{\theta_0 \in \mathbb{R}, \theta \in \Theta}{\arg\min} \sum_{i=1}^{N_{co}} \left(\theta_0 + \sum_{t=1}^{T_{pre}} \theta_t Y_{it} - \frac{1}{T_{\text{post}}} \sum_{t=T_{\text{pre}}+1}^{T} Y_{it} \right)^2 + \zeta^2 N_{\text{co}} \|\theta\|^2 \tag{2}$$

where the second term is a small regularization term to ensure uniqueness of time weights such that $\zeta = 1 \times 10^{-6} \widehat{\sigma}$ and

$$\Theta = \left\{ \theta \in \mathbb{R}_+^T, \text{ with } \sum_{t=1}^{T_{\text{pre}}} \theta_t = 1 \text{ and } \theta_t = \frac{1}{T_{\text{post}}} \text{ for all } t = T_{\text{pre}} + 1, \ldots, T \right\}$$

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Matrix Completion as a missing-data problem (1)

- For each unit $i \in [N]$ and time $t \in [T]$, we observe

$$Y_{it} = D_{it} Y_{it}(1) + (1 - D_{it}) Y_{it}(0), \qquad D_{it} \in \{0, 1\}.$$

- For treated cells with $D_{it} = 1$, the corresponding $Y_{it}(0)$ are *missing*. Our goal is to impute these missing entries to recover treatment effects.

- One can cast this causal inference problem with panel data as *missing-not-at-random* problem (Agarwal et al., 2021).

- It is natural to impute treated counterfactual when the ATT, if well defined, is the primary quantity of interest (Athey et al., 2021).

Matrix Completion as a missing-data problem (2)

- Stack the untreated potential outcomes into an $N \times T$ matrix $\mathbf{Y}(0)$. We only observe entries on a set of cells $\mathcal{O}$, with missing set $\mathcal{M}$.

$$\mathbf{Y}(0) \approx \begin{pmatrix} Y_{11}(0) & Y_{12}(0) & ? & \cdots & Y_{1T}(0) \\ ? & ? & Y_{23}(0) & \cdots & ? \\ Y_{31}(0) & ? & Y_{33}(0) & \cdots & ? \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ Y_{N1}(0) & ? & Y_{N3}(0) & \cdots & ? \end{pmatrix}, \qquad \mathbf{D} = \begin{pmatrix} 0 & 0 & 1 & \cdots & 0 \\ 1 & 1 & 0 & \cdots & 1 \\ 0 & 1 & 0 & \cdots & 1 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 1 & 0 & \cdots & 1 \end{pmatrix}$$

$$D_{it} = \begin{cases} 1 & \text{if } (i,t) \in \mathcal{M} \quad \text{(treated cells; } Y_{it}(0) \text{ missing)}, \\ 0 & \text{if } (i,t) \in \mathcal{O} \quad \text{(control cells; } Y_{it}(0) \text{ observed)}. \end{cases}$$

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Example: Neighborhood smoothing distance with paired observations (Y, X)

Setup. Three units ($i \in \{1, 2, 3\}$), three time periods, and $T_0 = 2$ pre-treatment periods. At each time $t \leq T_0$, we observe a pair

$$w_{i,t} = (Y_{i,t}, X_{i,t}) \in \mathbb{R}^2, \qquad W_{i,1:T_0} = \big(w_{i,1}, w_{i,2}\big).$$

Compute $\rho(1, 2)$ (here $k = 3$).

$$W_{1,1:2} - W_{2,1:2} = \big((0, 3) - (1, 2),\ (1, 1) - (0, 4)\big) = \big((-1, 1), (1, -3)\big).$$

$$\langle W_{3,1:2},\ W_{1,1:2} - W_{2,1:2} \rangle = (2, 0) \cdot (-1, 1) + (1, 2) \cdot (1, -3) = (-2) + (-5) = -7.$$

$$\rho(1, 2) = \frac{1}{2} \, | - 7| = 3.5.$$

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